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1 Scientific Notes: Spatial Structure Analysis of Energy Patterns

Objective: Investigate the spatial structure and dynamical characteristics of Energy Pattern 1 (EP1) and Energy Pattern 2 (EP2) cyclones during intensification phase.

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1.1 1. Introduction

1.1.1 1.1 Motivation

Following the identification of three distinct energy patterns in South Atlantic cyclones through PCA-based K-Means clustering (see `cluster_analysis_energy_patterns/SCIENTIFIC_NOTES.md`), this analysis investigates the **spatial structure** of the most energetically active patterns.

Energy Pattern Characteristics (from cluster analysis): - **EP1 (11.6%)**: Large barotropic and baroclinic conversions; **exports energy** to surroundings - **EP2 (25.6%)**: Moderate balanced conversions; **imports energy** from large-scale environment
- **EP3 (62.7%)**: Weak energetics representing typical “day-to-day” cyclones

Rationale for EP1 and EP2 focus:

EP3 cyclones exhibit weak energy budget activity and represent the climatological background. EP1 and EP2 cyclones, being energetically distinct and more intense, are the primary focus for understanding what structural characteristics differentiate high-impact systems from typical cyclones.

1.1.2 1.2 Research Questions

1. What are the spatial structures of baroclinic instability (EGR) in EP1 vs EP2?
2. How do upper-level (200 hPa) and low-level (850 hPa) PV anomalies differ between patterns?
3. What thermal advection patterns characterize each energy pattern?
4. How does near-surface moisture distribution and convergence differ?

Approach: Composite analysis of ERA5 reanalysis fields during the intensification phase of all EP1 (N=444) and EP2 (N=979) cyclones.

1.2 2. Dataset

1.2.1 2.1 Sample Composition

Energy Pattern	Cluster ID	Cases Analyzed	Percentage
EP1	0	444	11.6%
EP2	2	979	25.6%

Source: All EP1 and EP2 cyclones from cluster analysis with complete lifecycle phases.

Temporal Coverage: Intensification phase only (6-hourly timesteps during deepening period)

1.2.2 2.2 ERA5 Reanalysis Data

Variables Downloaded: - **Pressure levels (hPa):** 175, 200, 225, 250, 500, 825, 850, 875, 975

- **Pressure-level variables:** u, v, t, z, q (winds, temperature, geopotential, specific humidity)

- **Single-level variables:** msl (mean sea level pressure)

Spatial Configuration: - **Resolution:** $0.25^\circ \times 0.25^\circ$ - **Domain:** $30^\circ \times 30^\circ$ centered on cyclone track center - **Composite domain:** $15^\circ \times 15^\circ$ (marked in figures)

Temporal Resolution: 6-hourly

1.3 3. Methodology

1.3.1 3.1 Diagnostic Fields Computed

Diagnostic	Level(s)	Purpose	Key Reference
Eady Growth Rate	250–850 hPa	Baroclinic instability measure	Lindzen & Farrell (1980)
Potential Vorticity	200 hPa	Upper-level dynamics, tropopause folding	Hoskins et al. (1985)
Potential Vorticity	850 hPa	Low-level PV anomaly, diabatic effects	Čampa & Wernli (2012)
Temperature Advection	850 hPa	Warm/cold advection patterns	Sanders & Gyakum (1980)
Specific Humidity	975 hPa	Near-surface moisture distribution	-
Moisture Flux Divergence	975 hPa	Moisture convergence (convective potential)	Banacos & Schultz (2005)
Sea Level Pressure	Surface	Cyclone intensity and position	-

1.3.2 3.2 Spherical Grid Spacing

All horizontal derivatives account for Earth's spherical geometry:

Meridional spacing:

$$dy = R_{\oplus} \Delta\phi$$

Zonal spacing:

$$dx = R_{\oplus} \cos(\phi) \Delta\lambda$$

where $R_{\oplus} = 6.371 \times 10^6$ m, ϕ = latitude, λ = longitude.

Implementation: `compute_spherical_grid_spacing(lat_1d, lon_1d)` - Verifies coordinate monotonicity

- Correct gradient signs regardless of coordinate direction - Uses `numpy.gradient()` for non-uniform grids

1.3.3 3.3 Eady Growth Rate (EGR)

Formula:

$$\sigma_{EGR} = 0.31 \frac{|f|}{N} \left| \frac{\partial \vec{V}}{\partial z} \right|$$

where: - $f = 2\Omega \sin(\phi)$ = Coriolis parameter - $N^2 = \frac{g}{\theta_v} \frac{\partial \theta_v}{\partial z}$ = static stability (Brunt-Väisälä frequency) - $\left| \frac{\partial \vec{V}}{\partial z} \right|$ = vertical wind shear magnitude

Layer: 250–850 hPa (captures main tropospheric baroclinic zone)

Quality Control: - $N^2 > 10^{-6} \text{ s}^2$ (exclude statically unstable regions) - $|\phi| > 5^\circ$ (avoid equatorial singularity) - $\sigma_{EGR} < 5 \text{ day}^{-1}$ (cap unrealistic values)

1.3.4 3.4 Potential Vorticity

Formula:

$$PV = -g(\zeta_{\theta} + f) \frac{\partial \theta}{\partial p}$$

where ζ_{θ} = relative vorticity on isentropic surface, θ = potential temperature.

Levels: - **200 hPa:** Uses 175, 200, 225 hPa (upper-level tropopause dynamics) - **850 hPa:** Uses 825, 850, 875 hPa (low-level diabatic PV anomaly)

Units: $\text{K m}^2 \text{ kg}^{-1} \text{ s}^{-1}$ (SI), converted to PVU ($1 \text{ PVU} = 10^{-6} \text{ K m}^2 \text{ kg}^{-1} \text{ s}^{-1}$) in figures

1.3.5 3.5 Temperature Advection

Formula:

$$\text{advT} = -\vec{V} \cdot \nabla T = -\left(u \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial y}\right)$$

Level: 850 hPa

Sign Convention: Positive = warm air advection; Negative = cold air advection

Units: K h^{-1} (converted from K s^{-1})

1.3.6 3.6 Moisture Flux Divergence

Formula:

$$\nabla \cdot (q\vec{V}) = \frac{\partial(qu)}{\partial x} + \frac{\partial(qv)}{\partial y}$$

1.4 Level: 975 hPa (near-surface)

1.5 4. Results

Note: Statistical summaries presented below are computed from spatial composites. Physical interpretation of spatial patterns and EP1 vs EP2 differences is currently under analysis.

1.5.1 4.1 Summary Statistics

1.5.1.1 Eady Growth Rate (250–850 hPa)

Statistic	EP1	EP2
Mean $\pm$ SD (day ⁻¹)	{EP1_EGR_MEAN} $\pm$ {EP1_EGR_STD}	{EP2_EGR_MEAN} $\pm$ {EP2_EGR_STD}
Median (day ⁻¹)	{EP1_EGR_MEDIAN}	{EP2_EGR_MEDIAN}
Range (day ⁻¹)	[{EP1_EGR_MIN}, {EP1_EGR_MAX}]	[{EP2_EGR_MIN}, {EP2_EGR_MAX}]

1.5.1.2 Potential Vorticity at 200 hPa

Statistic	EP1	EP2
Mean (PVU)	{EP1_PV200_MEAN}	{EP2_PV200_MEAN}
Range (PVU)	[{EP1_PV200_MIN}, {EP1_PV200_MAX}]	[{EP2_PV200_MIN}, {EP2_PV200_MAX}]

1.5.1.3 Potential Vorticity at 850 hPa

Statistic	EP1	EP2
Mean (PVU)	{EP1_PV850_MEAN}	{EP2_PV850_MEAN}
Range (PVU)	[{EP1_PV850_MIN}, {EP1_PV850_MAX}]	[{EP2_PV850_MIN}, {EP2_PV850_MAX}]

1.5.1.4 Temperature Advection at 850 hPa

Statistic	EP1	EP2
Domain mean (K h ⁻¹)	{EP1_ADVT_MEAN}	{EP2_ADVT_MEAN}
Max warm advection (K h ⁻¹)	{EP1_ADVT_MAX_WARM}	{EP2_ADVT_MAX_WARM}
Max cold advection (K h ⁻¹)	{EP1_ADVT_MAX_COLD}	{EP2_ADVT_MAX_COLD}

1.5.1.5 Sea Level Pressure

Statistic	EP1	EP2
Minimum (hPa)	{EP1_SLP_MIN}	{EP2_SLP_MIN}
Maximum (hPa)	{EP1_SLP_MAX}	{EP2_SLP_MAX}

1.5.2 4.2 Spatial Composite Figures

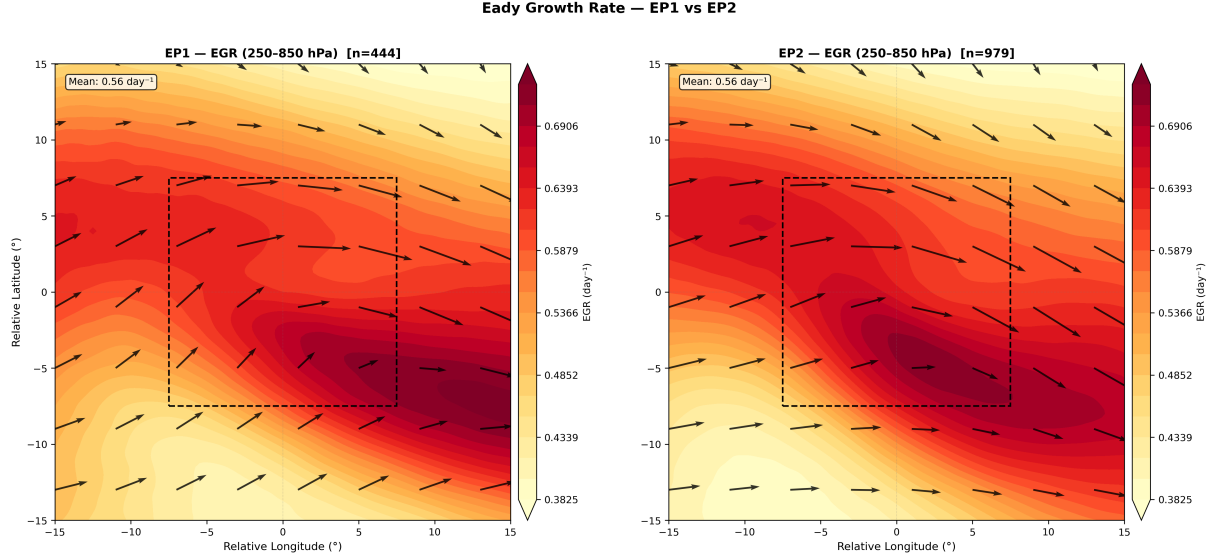


Figure 1: EGR Composite

1.5.2.1 Eady Growth Rate (Baroclinic Instability) *Figure: Eady growth rate composite (250–850 hPa layer) for EP1 (left) and EP2 (right). Contours show growth rate in day⁻¹. The 15° × 15° box marks the LEC computation domain.*

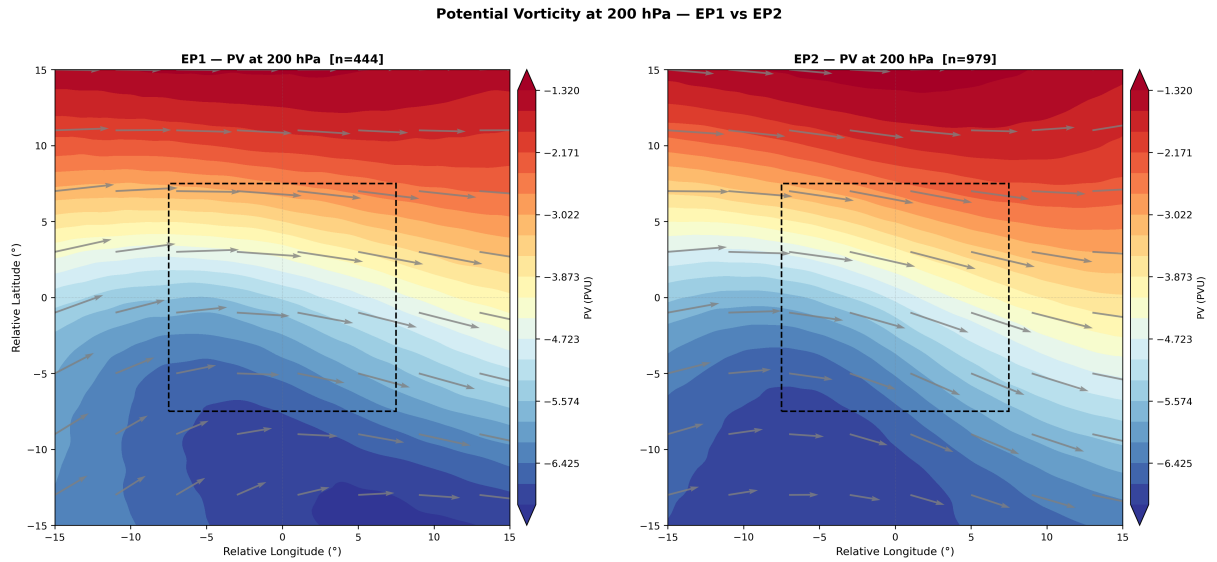


Figure 2: PV@200 Composite

1.5.2.2 Upper-Level Dynamics (200 hPa PV) *Figure: Potential vorticity at 200 hPa for EP1 (left) and EP2 (right). Units: PVU (10⁻⁶ K m² kg⁻¹ s⁻¹). Shows tropopause dynamics and stratospheric intrusions.*

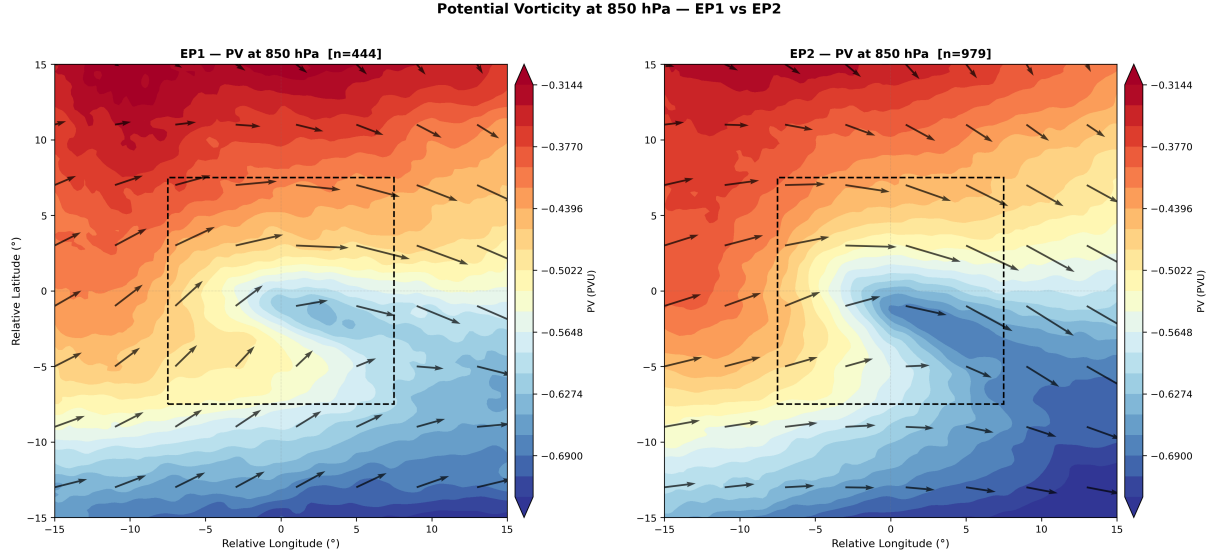


Figure 3: PV@850 Composite

1.5.2.3 Low-Level PV Anomaly (850 hPa) *Figure: Potential vorticity at 850 hPa for EP1 (left) and EP2 (right). Units: PVU. Indicates diabatic PV generation and low-level cyclonic circulation.*

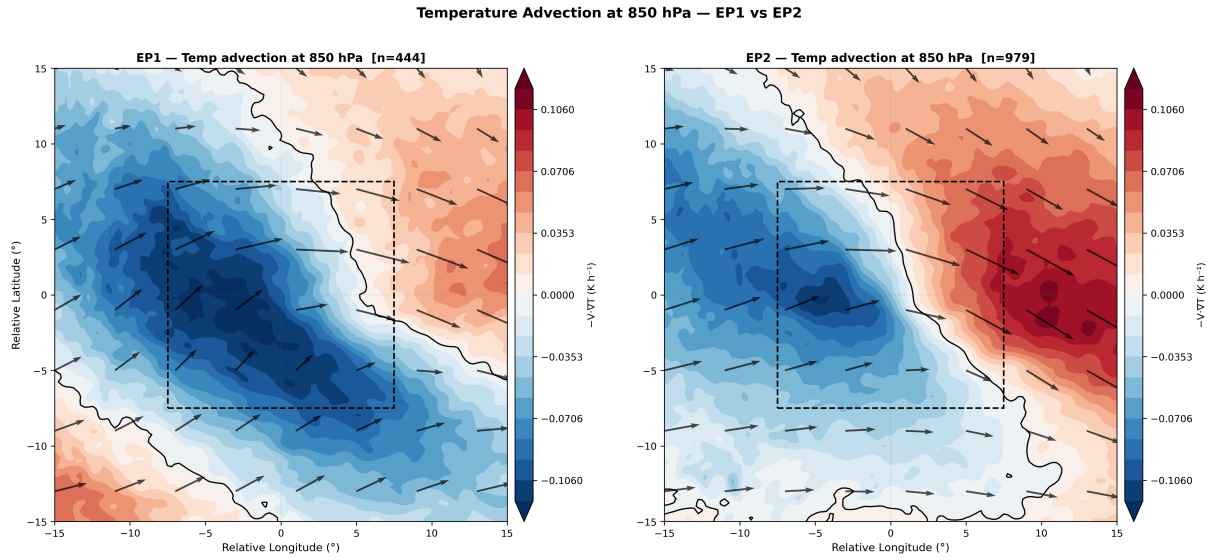


Figure 4: Temp Advection Composite

1.5.2.4 Temperature Advection Pattern *Figure: Temperature advection at 850 hPa for EP1 (left) and EP2 (right). Units: $K h^{-1}$. Positive (red) = warm advection; Negative (blue) = cold advection.*

1.5.2.5 Moisture Distribution and Convergence *Figure: Specific humidity (shading, $g kg^{-1}$) and moisture flux divergence (contours, $g kg^{-1} s^{-1}$) at 975 hPa for EP1 (left) and EP2 (right). Negative divergence (dashed contours) indicates moisture convergence.*

1.5.2.6 Sea Level Pressure *Figure: Mean sea level pressure for EP1 (left) and EP2 (right). Units: hPa. Shows cyclone intensity and horizontal structure.*

Moisture Composite

Figure 5: Moisture Composite

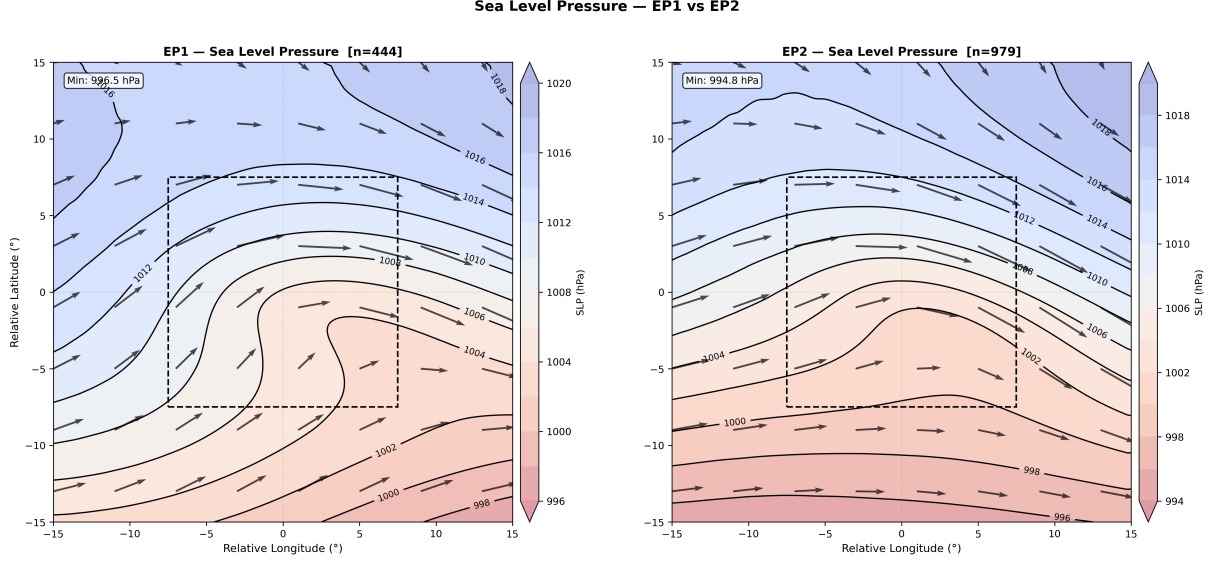


Figure 6: SLP Composite

1.6 5. Physical Interpretation

Status: Analysis of spatial patterns and physical mechanisms is ongoing. Preliminary observations include:

1.6.1 5.1 EP1 Characteristics

- {EP1_INTERPRETATION}

1.6.2 5.2 EP2 Characteristics

- {EP2_INTERPRETATION}

1.6.3 5.3 Key Structural Differences

- {KEY_DIFFERENCES}

Future Work: Detailed analysis of spatial patterns, vertical coupling mechanisms, and relationship between energetics (from cluster analysis) and structural characteristics.

1.7 6. Computational Implementation

1.7.1 6.1 Grid Spacing on Spherical Earth

Function: `compute_spherical_grid_spacing(lat_1d, lon_1d)`

Computes accurate grid spacing accounting for Earth's spherical geometry:

```
# Meridional spacing (constant per latitude band)
```

```
dy = R_EARTH * Δ # meters
```

```
# Zonal spacing (varies with latitude)
```

```
dx = R_EARTH * cos( ) * Δ # meters
```

Quality checks: - Verifies latitude is monotonic (increasing or decreasing) - Verifies longitude is monotonic - Raises `ValueError` if non-monotonic - Gradient sign automatically correct regardless of coordinate direction

1.7.2 6.2 Divergence on Spherical Coordinates

For small domains ($\sim 30^\circ$), the simplified Cartesian approximation is valid:

$$\nabla \cdot \vec{F} \approx \frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y}$$

where dx and dy account for spherical geometry (Section 6.1).

Full spherical formula (not required for 30° domain):

$$\nabla \cdot \vec{F} = \frac{1}{R \cos \phi} \frac{\partial F_\lambda}{\partial \lambda} + \frac{1}{R \cos \phi} \frac{\partial (F_\phi \cos \phi)}{\partial \phi}$$

Our implementation uses the simplified form with accurate dx , dy — appropriate for mesoscale domains.

1.7.3 6.3 Unit Tracking

All calculations use **MetPy units** for automatic dimensional analysis:

```
# Example: moisture flux
```

```
qu = (q * units('kg/kg')) * (u * units('m/s'))
```

```
# → qu.units = 'kg / kg * m / s' (automatically tracked)
```

```
# Convert divergence to g/kg/s
```

```
div_q_si = (dqu_dx + dqv_dy) * units('kg/kg/s')
```

```
div_q_gkg = (div_q_si * 1000 * units('g/kg')).magnitude
```

Benefits: - Prevents unit errors (e.g., mixing K and $^\circ\text{C}$) - Documents unit transformations
- Explicit conversions (no hardcoded magic numbers)

1.7.4 6.4 Numerical Methods

- **Spatial derivatives:** `numpy.gradient()` with 2nd-order centred finite differences
- **Vertical derivatives:** Centred FD over 3 levels (e.g., 175–200–225 hPa for PV@200)
- **Interpolation:** Linear interpolation to regular 0.25° grid via `xarray.interp()`

1.7.5 6.5 Quality Control Filters

Diagnostic	Filter	Rationale
EGR	$N^2 > 10^{-6} \text{ s}^{-2}$	Exclude statically unstable regions
EGR	$\ \phi\ > 5^\circ$	Avoid near-equatorial ($f \approx 0$)
EGR	$\sigma_{EGR} < 5 \text{ day}^{-1}$	Cap unrealistic growth rates

Diagnostic	Filter	Rationale
PV	Valid only where θ monotonic	Ensure isentropic sorting

1.7.6 6.6 Code Verification

Test cases: 1. Zonal wind $\rightarrow$ zero meridional derivative 2. Meridional wind $\rightarrow$ zero zonal derivative

3. Uniform field $\rightarrow$ zero divergence 4. Reversed latitude ($90 \rightarrow -90$) $\rightarrow$ correct gradient sign

Validation: - Cross-check EGR values with literature (typical: $0.3\text{--}1.0 \text{ day}^{-1}$) - PV@200 should align with tropopause ($\sim 2\text{--}3$ PVU contour) - Compare temperature advection with operational analyses

1.8 7. References

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